A FINITE MORPHISM IS PSEUDO-COHERENT EXACTLY WHEN THE PUSHFORWARD STRUCTURE SHEAF IS A PSEUDO-COHERENT MODULE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17454-17459 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17461-17470, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The morphism-level avatar of the algebra lemma also powering L16133; counted separately per distinct aggregate (module-vs-relative equivalence for the structure sheaf, versus complex pushforward invariance). The pool entry is lemma-finite-pseudo-coherent at lines 17454-17470.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of pseudo-coherent morphism (chapter definition section, via polynomial presentation relative pseudo-coherence of O_X); More on Algebra, lemma finite-extension-pseudo-coherent (pseudo-coherence over a finite extension is independent of the side).

Source reference. Original lemma 0673: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17470–17470. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a finite morphism of schemes. Then f is pseudo-coherent if and only if $f_*\mathcal{O}_X$ is pseudo-coherent as an $\mathcal{O}_S$-module.*

Proof. Translated into algebra this lemma says the following: If $R \to A$ is a finite ring map, then $R \to A$ is pseudo-coherent as a ring map (which means by definition that A as an A-module is pseudo-coherent relative to R) if and only if A is pseudo-coherent as an R-module. This follows from the more general More on Algebra, Lemma 0673. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.